

CHANY, Russian Federation

WMO: 296020

Lat: 55.3168N

Lon: 76.7445E

Elev: 111

StdP: 100.00

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.2	-33.0	-39.2	0.1	-36.1	-36.0	0.1	-32.8	9.2	-9.6	8.2	-9.9	1.7	340	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	29.6	18.8	28.0	18.4	26.4	17.6	20.6	26.3	19.6	25.4	18.7	24.4	4.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.6	23.0	17.6	12.7	22.0	16.6	12.0	21.3	60.0	26.2	56.7	25.6	53.8	24.4	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.1	7.2	-37.9	32.3	4.6	2.0	-41.2	33.7	-43.9	34.9	-46.5	36.0	-49.8	37.4		
(5)			-38.0	22.1	4.4	1.2	-41.1	23.0	-43.7	23.7	-46.1	24.4	-49.3	25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-18.8	-16.2	-6.7	4.2	11.8	17.3	19.0	16.4	10.5	3.0	-7.2	-14.9		
	DBStd	14.57	8.25	7.46	5.91	5.88	5.48	4.38	3.45	3.84	4.69	4.94	7.79	8.18		
	HDD10.0	3947	894	733	518	191	46	4	0	2	49	222	516	772		
	HDD18.3	6231	1153	967	776	426	211	70	36	84	239	474	766	1030		
	CDD10.0	887	0	0	0	16	103	222	278	199	64	6	0	0		
	CDD18.3	130	0	0	0	0	9	38	55	24	4	0	0	0		
	CDH23.3	1249	0	0	0	3	135	363	472	234	43	0	0	0		
	CDH26.7	296	0	0	0	0	29	93	114	52	8	0	0	0		
(14) Wind	WSAvg	3.4	3.1	3.2	3.7	3.9	3.9	3.2	3.0	3.1	3.3	3.4	3.6	3.4		
(15) Precipitation	PrecAvg	381	18	16	15	18	25	53	66	52	34	25	33	26		
	PrecMax	553	38	27	39	42	51	123	150	98	74	60	75	46		
	PrecMin	273	2	4	1	2	9	16	18	10	10	8	6	11		
	PrecStd	79	10	6	11	11	14	25	34	20	19	15	19	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.3	0.9	6.4	22.8	30.0	31.2	31.8	30.6	27.4	18.9	7.1	1.3		
		MCWB	-0.1	-0.1	3.3	13.0	16.5	18.8	20.3	19.0	15.8	11.0	5.3	0.5		
	2%	DB	-2.0	-0.9	4.0	18.9	26.9	29.3	29.4	28.0	24.1	15.8	5.4	0.1		
		MCWB	-2.6	-1.7	1.9	11.3	15.9	18.2	19.6	18.6	15.0	9.6	4.1	-0.4		
	5%	DB	-4.2	-2.9	2.5	16.3	24.5	27.4	27.8	26.0	21.1	12.9	3.4	-1.8		
		MCWB	-4.7	-3.5	1.0	10.2	14.6	17.8	19.1	17.3	13.8	8.2	2.3	-2.4		
	10%	DB	-6.5	-5.2	1.2	13.2	21.7	25.4	26.1	23.9	18.3	10.2	1.4	-3.9		
		MCWB	-7.0	-5.8	0.0	8.1	13.4	17.2	18.6	16.9	12.3	6.9	0.5	-4.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.0	0.2	3.7	14.0	18.4	21.2	22.1	20.8	17.4	12.0	6.0	0.9		
		MCDB	0.3	0.5	5.6	20.4	27.1	27.8	28.6	27.0	24.8	16.9	6.5	1.1		
	2%	WB	-2.6	-1.6	2.1	12.3	16.7	19.7	21.0	19.5	15.7	10.3	4.1	-0.5		
		MCDB	-2.1	-0.9	3.6	18.0	25.2	26.0	26.5	25.7	22.1	14.9	5.3	0.1		
	5%	WB	-4.6	-3.5	1.2	10.3	15.3	18.8	20.2	18.5	14.4	8.6	2.4	-2.3		
		MCDB	-4.2	-2.8	2.4	15.6	22.9	25.0	25.7	24.1	19.9	12.3	3.4	-1.8		
	10%	WB	-7.0	-5.7	0.1	8.6	13.9	17.8	19.2	17.4	12.9	7.1	0.6	-4.5		
		MCDB	-6.6	-5.2	1.3	13.0	20.6	23.7	24.6	22.5	17.8	10.0	1.2	-3.9		
(35) Mean Daily Temperature Range	MDBR		7.7	8.4	8.6	9.1	11.3	10.5	10.2	10.5	9.9	7.4	6.1	7.5		
	5% DB	MCDBR	8.0	7.3	7.7	13.5	15.0	13.5	13.0	14.0	14.0	11.9	6.1	7.9		
		MCWBR	7.8	7.1	6.1	7.7	6.5	5.2	4.9	5.8	6.4	7.0	5.1	7.8		
	5% WB	MCDBR	7.8	7.3	6.8	12.7	13.8	11.7	11.2	11.7	12.7	10.7	5.8	8.1		
		MCWBR	7.7	7.1	5.7	7.6	6.9	5.2	5.0	5.5	6.4	6.7	5.1	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.316	0.374	0.406	0.401	0.409	0.431	0.419	0.365	0.352	0.320	0.287		
	taud		2.033	2.004	1.926	2.112	2.233	2.286	2.226	2.228	2.340	2.229	2.079	1.983		
	Ebn at Noon		599	723	763	808	841	837	807	790	790	687	543	483		
	Edh at Noon		68	102	141	136	128	123	129	121	94	82	65	57		
(44) All-Sky Solar Radiation	RadAvg		0.95	2.03	3.62	4.63	5.47	5.92	5.42	4.39	3.10	1.54	0.74	0.61		
	RadStd		0.07	0.10	0.20	0.39	0.49	0.51	0.37	0.42	0.30	0.18	0.12	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page